

Solution: a) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a strictly decreasing function.

By Defn. 6, $\forall x \neq y, x, y \in \mathbb{R}$ and $x < y, f(x) > f(y)$.

Let, $\exists a, b \in \mathbb{R}$ s.t. $f(a) = f(b)$. If $a > b$, then by Defn. 6, $f(a) < f(b)$.

If $a < b$, then by Defn. 6, $f(a) > f(b)$. $\therefore a = b$.

By Defn. 5, f is one-to-one. $\square$

b) Give an example of a decreasing function from $\mathbb{R}$ to itself that is not one-to-one.

Solution: $f(x) = 1/\lfloor x \rfloor$. By Prob 26(b), we know that the function $g(x) = \lfloor x \rfloor$ is an increasing function. By Prob 24 and some little tweaking of arguments, $f(x)$ is a decreasing function.

$$f(1.1) = 1/\lfloor 1.1 \rfloor = 1/1 = 1 \quad f(1.2) = 1/\lfloor 1.2 \rfloor = 1/1 = 1$$

We know, $1.1, 1.2 \in \mathbb{R}$ but $1.1 \neq 1.2$. $\therefore$ By Defn 5, f is not one-to-one. $\square$

28) Show that the function $f(x) = e^x$ from the set of real numbers to the set of real numbers is not invertible, but if the codomain is restricted to the set of positive real numbers, the resulting function is invertible.

Solution: $f: \mathbb{R} \rightarrow \mathbb{R}$. Let, $f(x) = -1$. $\therefore e^x = -1 \Rightarrow x = \ln(-1)$. We know $\ln(-1)$ does not exist. By Defn. 7, f is not onto. By Defn. 8, f is not a bijection. By Defn. 9, f has no inverse.

$$f: \mathbb{R} \rightarrow \mathbb{R}^+. \text{ Let, } x, y \in \mathbb{R} \text{ and } f(x) = f(y). \therefore e^x = e^y \Rightarrow \frac{e^x}{e^y} = 1$$

$$\Rightarrow e^{x-y} = e^0 \Rightarrow x-y=0 \Rightarrow x=y. \text{ By Defn. 5, } f \text{ is one-to-one.}$$

$$\text{Let, } f(x) = y, \text{ where } y \in \mathbb{R}^+. \therefore e^x = y \Rightarrow x = \ln(y). \text{ We know } \ln(y) \in \mathbb{R}.$$

$$\therefore \forall y \in \mathbb{R}^+, \exists x \in \mathbb{R} \text{ s.t. } f(x) = y. \text{ By Defn. 7, } f \text{ is onto.}$$

By Defn. 8, f is a bijection. By Defn. 9, f is invertible.

29) Show that the function $f(x) = |x|$ from the set of real numbers to the set of nonnegative real numbers is not invertible, but if the domain is restricted to the set of non-negative real numbers, the resulting function is invertible.